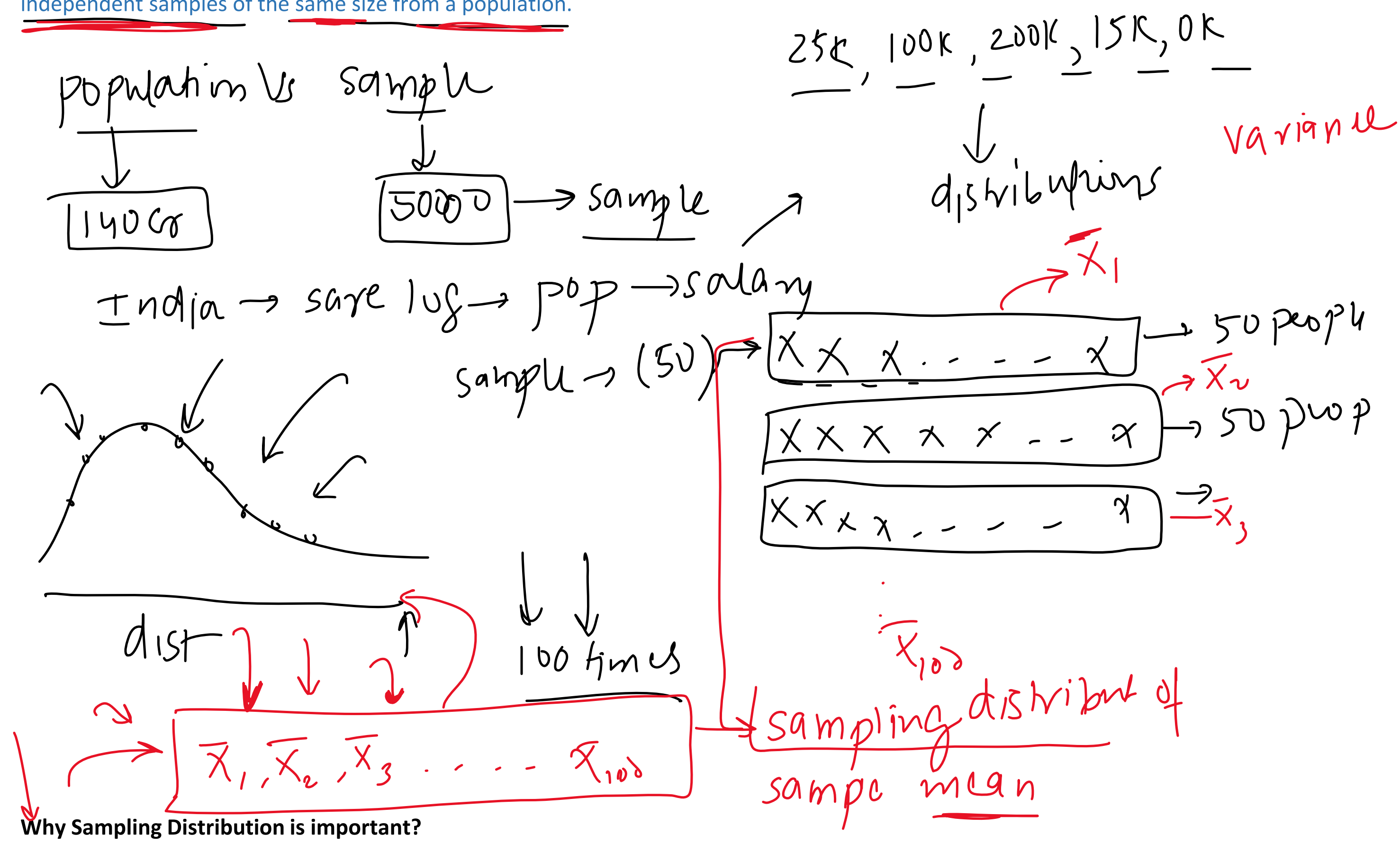
Sampling Distribution

27 March 2023 17:10

Sampling distribution is a probability distribution that describes the statistical properties of a sample statistic (such as the sample mean or sample proportion) computed from multiple independent samples of the same size from a population.



Why Sampling Distribution is important?

Sampling distribution is important in statistics and machine learning because it allows us to estimate the variability of a sample statistic, which is useful for making inferences about the population. By analysing the properties of the sampling distribution, we can compute confidence intervals, perform hypothesis tests, and make predictions about the population based on the sample data.

→ central limit theorem

Central Limit Theorem

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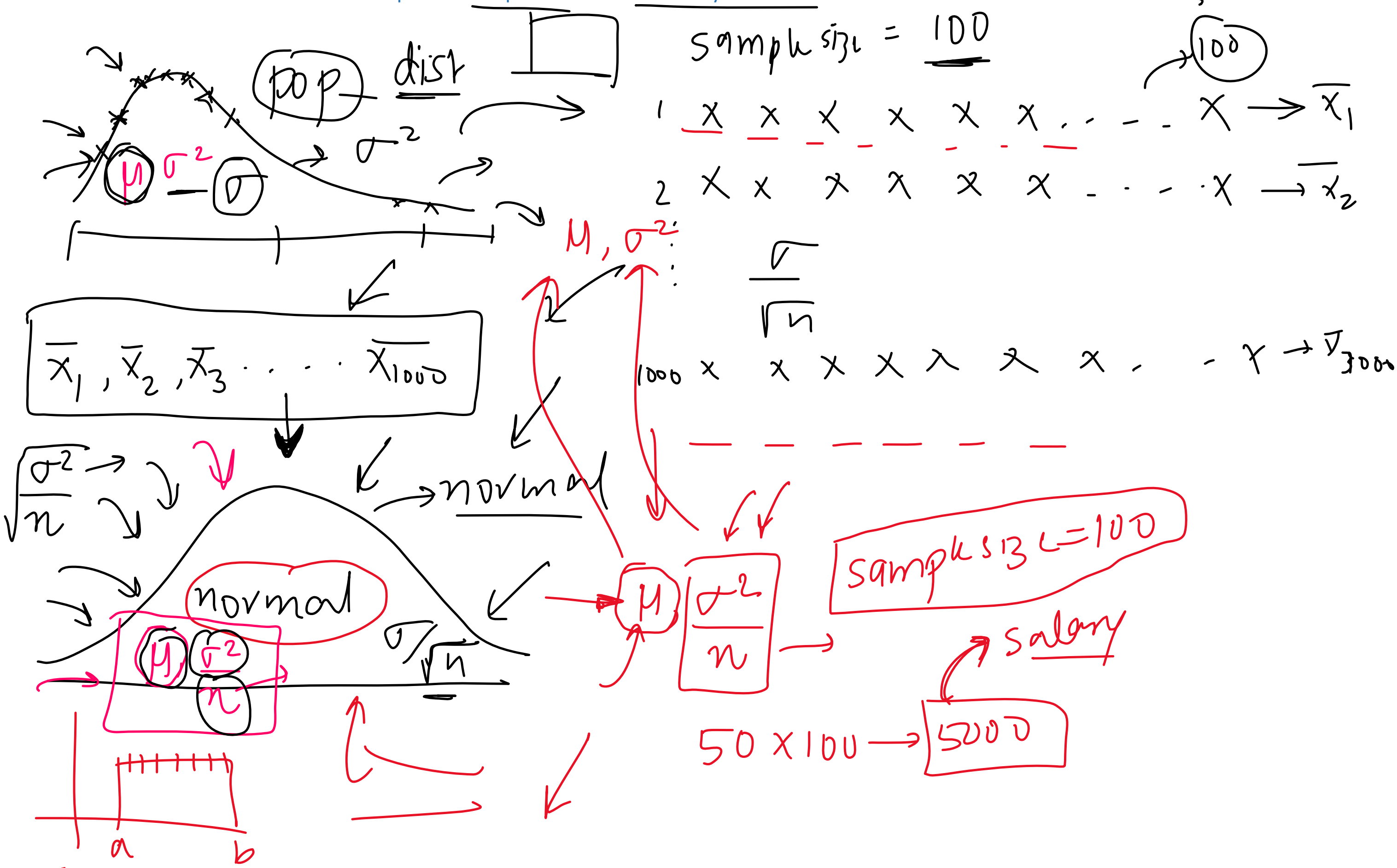
interval $\boxed{\mu = 32.584}$ fare $\mu = 32.584$

The Central Limit Theorem (CLT) states that the distribution of the sample means of a large number of independent and identically distributed random variables will approach a normal distribution, regardless of the underlying distribution of the variables.

The conditions required for the CLT to hold are:

1. The sample size is large enough, typically greater than or equal to 30.
2. The sample is drawn from a finite population or an infinite population with a finite variance.
3. The random variables in the sample are independent and identically distributed.



The CLT is important in statistics and machine learning because it allows us to make probabilistic inferences about a population based on a sample of data. For example, we can use the CLT to construct confidence intervals, perform hypothesis tests, and make predictions about the population mean based on the sample data. The CLT also provides a theoretical justification for many commonly used statistical techniques, such as t-tests, ANOVA, and linear regression.

Case Study 1 - Titanic Fare

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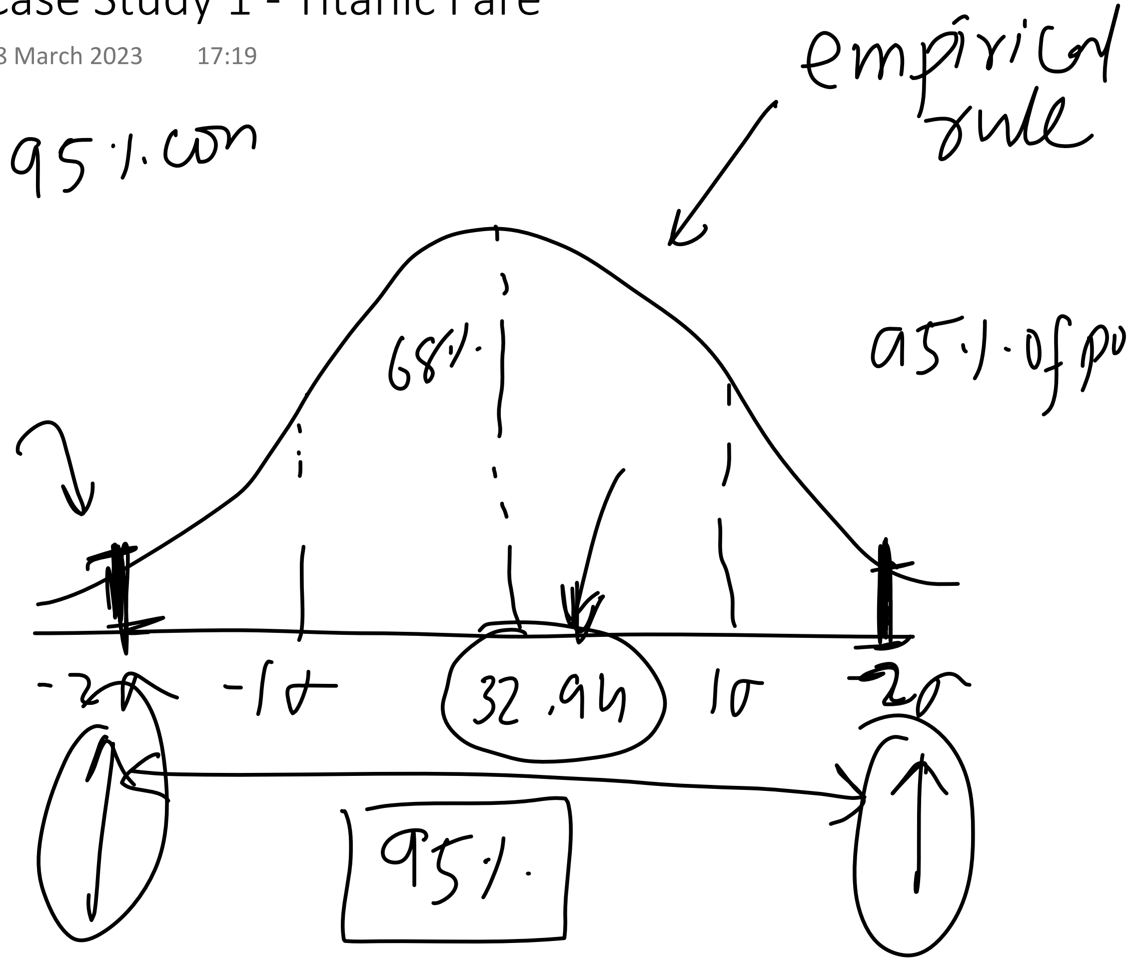

Case Study - What is the average income of Indians

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Step-by-step process:

1. Collect multiple random samples of salaries from a representative group of Indians. Each sample should be large enough (usually, n > 30) to ensure the CLT holds. Make sure the samples are representative and unbiased to avoid skewed results.

2. Calculate the sample mean (average salary) and sample standard deviation for each sample.

3. Calculate the average of the sample means. This value will be your best estimate of the population mean (average salary of all Indians).

4. Calculate the standard error of the sample means, which is the standard deviation of the sample means divided by the square root of the number of samples.

5. Calculate the confidence interval around the average of the sample means to get a range within which the true population mean likely falls. For a 95% confidence interval:

   lower_limit = average_sample_means - 1.96 * standard_error
   upper_limit = average_sample_means + 1.96 * standard_error

6. Report the estimated average salary and the confidence interval.

Python code

Remember that the validity of your results depends on the quality of your data and the representativeness of your samples. To obtain accurate results, it's crucial to ensure that your samples are unbiased and representative.